

Mikael GALLEGO

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EDUCATION

Université de Bordeaux | *Master Physique fondamentale et applications* 2024 — Present
Specialization: Light, Matter, and iNteractions (LMN)
North Carolina State University | *B.S. Physics, B.S. Mathematics* 2020 — 2024
Magna Cum Laude

COURSEWORK

- Quantum Mechanics, Quantum Optics
- Laser Physics and Nonlinear Optics
- Electrodynamics
- Statistical Mechanics
- Optics of Nanomaterials
- Nanophysics, Nanophotonics
- Numerical Analysis, Physics Simulations
- Real and Complex Analysis, Group Theory

SKILLS

- **Computer languages and software:** C, Python, Wolfram, L^AT_EX, Microsoft Office, Linux/UNIX, OSLO, MPI, IBM Spectrum LSF, LabView.
- **Human languages:** English (native), French (C1).

EXPERIENCE

Laboratoire Ondes et Matière d'Aquitaine (LOMA) | *Intern* Feb. 2026 — Present
• Apply time-frequency analysis methods like the wavelet transform to real human EEG data to detect and characterize sleep spindles. Find statistical correlations between patients. Work is theoretical and computational.

Integration from Material to System Laboratory (IMS) | *Intern* Jan. 2025 — July 2025
• Designed and built a pump-probe setup for an array of terahertz-emitting nanoantennae to be used for non-destructive sensing. Work was experimental, with a computational aspect involving modeling mirrors.

Nonlinear Artificial Intelligence Lab (NAIL) | *Research Assistant* Feb. 2023 — Feb. 2026
• Investigated the effect of spatial disorder ‘taming’ chaotic systems of nonlinear pendula. Extended previous work to parameter limits via computing cluster. Work was computational. Resulted in a publication.

NCSU Physics Senior Laboratory | *Student* Aug. 2023 — Dec. 2023
• Performed coincidence-detection experiments on correlated photon pairs. Work was experimental.

NCSU Physics Department | *Employee* April 2022 — July 2024
• Constructed classroom demonstrations for physics and engineering professors. Work was experimental.

PUBLICATION

- **M. Gallego**, J. F. Lindner, A. Radhakrishnan, W. L. Ditto, Lyapunov landscape elucidates disorder-taming-chaos in arrays of coupled pendula, *Physics Letters A* 579 (2026) 131483.
DOI: 10.1016/j.physleta.2026.131483

CONFERENCES

- Presented: “Photoconductive terahertz antenna source array for nondestructive testing”, EUR LIGHT Master Workshop, Université de Bordeaux, June 2025.
- Presented: “Disorder Taming Chaos in Coupled Pendulum Arrays”, Undergraduate Research Symposium, North Carolina State University, April 2024.
- Presented: “Single Photon Quantum Mechanics”, Senior Physics Lab Presentations, North Carolina State University, December 2023.